

## HIGHLAND HOME 2 S, AL, USA

WMO: 722266

Lat: 31.9155N

Lon: 86.3115W

Elev: 187

StdP: 99.10

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 73802

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |            |            |            |            | Coldest Month WS/MCDB |            |            |            | MCWS/PCWD to 99.6% DB |            | WSF        |
|---------------|-------------|-------------|-------------------------------|------------|------------|------------|------------|------------|-----------------------|------------|------------|------------|-----------------------|------------|------------|
|               |             |             | 99.6%                         |            |            | 99%        |            |            | 0.4%                  |            | 1%         |            |                       |            |            |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB       | DP         | HR         | MCDB       | WS                    | MCDB       | WS         | MCDB       | MCWS                  | PCWD       |            |
| (a)<br>1      | (b)<br>-4.4 | (c)<br>-2.1 | (d)<br>N/A                    | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A            | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A            | (o)<br>N/A | (p)<br>N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |            |             |            |             |            | Evaporation WB/MCDB |            |            |            |            |            | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|------------|-------------|------------|-------------|------------|---------------------|------------|------------|------------|------------|------------|----------------------|------------|
|               |                        | 0.4%            |            | 1%          |            | 2%          |            | 0.4%                |            | 1%         |            | 2%         |            |                      |            |
|               |                        | DB              | MCWB       | DB          | MCWB       | DB          | MCWB       | WB                  | MCDB       | WB         | MCDB       | WB         | MCDB       | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>9.4             | (c)<br>34.1     | (d)<br>N/A | (e)<br>33.0 | (f)<br>N/A | (g)<br>32.0 | (h)<br>N/A | (i)<br>N/A          | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>N/A           | (p)<br>N/A |

| Dehumidification DP/MCDB and HR |            |            |            |            |            |            |            |            | Enthalpy/MCDB |            |            |            |            |            | Extreme<br>Max WB |
|---------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|---------------|------------|------------|------------|------------|------------|-------------------|
| 0.4%                            |            |            | 1%         |            |            | 2%         |            |            | 0.4%          |            | 1%         |            | 2%         |            |                   |
| DP                              | HR         | MCDB       | DP         | HR         | MCDB       | DP         | HR         | MCDB       | Enth          | MCDB       | Enth       | MCDB       | Enth       | MCDB       |                   |
| (a)<br>N/A                      | (b)<br>N/A | (c)<br>N/A | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A    | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>N/A |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| N/A               | N/A  | N/A | -8.0                       | 35.2 | 2.6                | 1.9 | -9.8                                               | 36.6 | -11.3      | 37.7 | -12.8      | 38.8 | -14.6      | 40.2 |
|                   |      |     | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |
|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |
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|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |
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|                   |      |     |                            |      |                    | </  |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 18.5   | 8.4   | 11.3  | 14.5  | 18.4  | 22.3  | 25.8  | 26.5  | 26.4  | 24.7  | 19.5  | 13.4  | 10.8  |
|                                                                        | DBStd       | 7.54   | 5.91  | 5.52  | 4.91  | 3.33  | 3.02  | 1.91  | 1.60  | 1.72  | 2.67  | 4.04  | 4.59  | 5.37  |
|                                                                        | HDD10.0     | 250    | 102   | 48    | 19    | 1     | 0     | 0     | 0     | 0     | 0     | 1     | 19    | 59    |
|                                                                        | HDD18.3     | 1125   | 311   | 202   | 133   | 40    | 8     | 0     | 0     | 0     | 1     | 35    | 157   | 238   |
|                                                                        | CDD10.0     | 3367   | 52    | 85    | 159   | 252   | 382   | 473   | 513   | 510   | 439   | 297   | 122   | 84    |
|                                                                        | CDD18.3     | 1201   | 2     | 5     | 16    | 41    | 132   | 223   | 254   | 252   | 191   | 73    | 9     | 4     |
|                                                                        | CDH23.3     | 9751   | 1     | 14    | 76    | 282   | 976   | 1989  | 2236  | 2125  | 1526  | 488   | 34    | 6     |
|                                                                        | CDH26.7     | 3652   | 0     | 0     | 4     | 31    | 281   | 816   | 929   | 885   | 587   | 116   | 3     | 0     |
| (14) Wind                                                              | WSAvg       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (15) Precipitation                                                     | PrecAvg     | 1401   | 127   | 131   | 167   | 112   | 102   | 124   | 128   | 116   | 104   | 73    | 112   | 114   |
|                                                                        | PrecMax     | 2230   | 266   | 324   | 359   | 356   | 244   | 328   | 280   | 242   | 335   | 308   | 302   | 249   |
|                                                                        | PrecMin     | 1071   | 23    | 30    | 13    | 11    | 3     | 31    | 30    | 23    | 13    | 0     | 18    | 40    |
|                                                                        | PrecStd     | 236    | 52    | 69    | 81    | 73    | 57    | 75    | 62    | 49    | 75    | 62    | 61    | 49    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 22.9  | 25.4  | 27.4  | 28.8  | 32.4  | 35.2  | 35.0  | 35.5  | 34.8  | 31.6  | 26.5  | 24.1  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 2%          | DB     | 20.8  | 22.9  | 25.5  | 27.7  | 30.9  | 33.7  | 33.5  | 33.8  | 32.9  | 29.5  | 24.0  | 22.2  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5%          | DB     | 19.0  | 21.1  | 23.8  | 26.5  | 29.6  | 32.3  | 32.3  | 32.5  | 31.6  | 27.8  | 22.4  | 20.6  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 10%         | DB     | 17.1  | 19.5  | 21.9  | 25.1  | 28.3  | 31.0  | 31.3  | 31.2  | 30.3  | 26.2  | 21.0  | 19.1  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 10.0  | 10.3  | 10.5  | 10.9  | 10.1  | 9.6   | 9.4   | 9.5   | 9.9   | 10.7  | 10.9  | 9.2   |
|                                                                        |             | MCDBR  | 10.0  | 10.3  | 11.1  | 11.9  | 11.3  | 11.2  | 10.5  | 11.2  | 11.5  | 11.4  | 11.1  | 9.5   |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.318 | 0.327 | 0.352 | 0.379 | 0.426 | 0.465 | 0.521 | 0.505 | 0.425 | 0.366 | 0.340 | 0.322 |
|                                                                        | taud        |        | 2.537 | 2.517 | 2.445 | 2.367 | 2.273 | 2.220 | 2.069 | 2.112 | 2.318 | 2.462 | 2.477 | 2.539 |
|                                                                        | Ebn at Noon |        | 909   | 934   | 929   | 912   | 866   | 828   | 780   | 788   | 844   | 877   | 872   | 885   |
|                                                                        | Edh at Noon |        | 84    | 94    | 109   | 123   | 136   | 143   | 165   | 157   | 123   | 99    | 89    | 80    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.86  | 3.43  | 4.62  | 5.72  | 6.27  | 6.13  | 5.96  | 5.53  | 4.95  | 4.25  | 3.23  | 2.47  |
|                                                                        | RadStd      |        | 0.19  | 0.43  | 0.41  | 0.45  | 0.43  | 0.48  | 0.39  | 0.33  | 0.39  | 0.51  | 0.29  | 0.27  |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (48 neighbors) | +0.45 | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +156    | +100    |

Nomenclature: See separate page